BLACKBOX

The ultimate game of hide and seek

Player's Guide

Introduction

BlackBox™ is a Parker Brothers (now owned by Hasbro) game, introduced in the late 1970s in which the player must deduce the location of hidden elements within a black box by applying inputs and observing the behavior of the outputs.

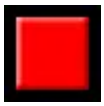
In the game, 3 to 5 balls (depending on how many the player chooses when starting the game) are hidden in an 8-by-8 grid. Inputs are supplied in the form of *rays* that the player *shoots* into the sides of the grid. Because these rays behave according to deterministic rules when in the vicinity of a ball within the grid, the locations of the balls can be deduced by observing the input and output characteristics of the rays.

The object of the game is to accurately guess the location of the hidden balls using the least amount of information.

Shooting Rays

The player shoots a ray into the grid by clicking on one of the gray **entry ports** (shown left) along the periphery of the grid. The port is replaced with a marker which indicates the result of the ray.

This result will be one of the following:



A **hit** is indicated by a red marker as shown left. A hit occurs when the ray enters the grid, but never exits, and is always the result of a ray directly hitting a ball head on, in which case the ray is *absorbed* by the ball.

A **reflection** is indicated by a yellow marker, shown left. A reflection occurs when the ray exits the grid at the same port through which it entered. This can be caused by a number of different situations which will be described below when we get to the rules of ray behavior.

A **detour** occurs when a ray exits the grid from a different port than it entered. It is indicated by a pair of orange marker with a symbol on them. The various symbols are not themselves significant, but are used to match the entry and exit ports. For example, the matching entry and exit port for one detoured ray could be shown with a triangle symbol, and the matching entry and exit ports for another ray could be shown with a circle symbol.

Detour markers are *always* placed in pairs using matching symbols.

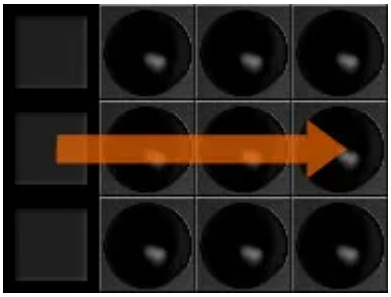
Additionally, it should be noted that (because of the way that rays behave) entry and exit points are commutative. If a ray shot into one port exits at a second, a ray shot into the second port would have emerged at the first.

Ray Behavior Rules

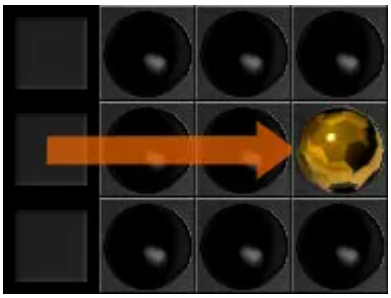
Rays behave using a simple set of ironclad rules. By logically applying these rules, the player must deduce the location of the hidden balls by observing the entry and exit behavior of the rays. The player cannot*see* what the rays do inside the grid — that's the intrinsic concept of a black box. The observed input/output behaviors must be used to deduce the internal action of the rays, and subsequently the location of the hidden balls.

The rules are presented below in order of strongest to weakest. In other words, Rule 1 takes precedence over Rule 2 and so on.

Rule 1: When no balls are in the vicinity, a ray will simply travel in a straight line. So if there are no balls along its path, a ray will simply travel straight through the grid and come out directly on the other side.

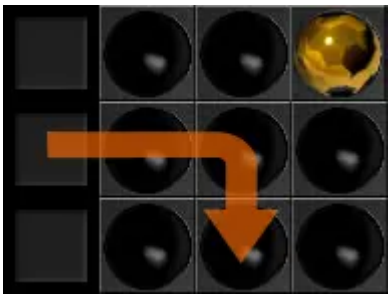


Rule 2: If a ray hits a ball straight on, it is absorbed by the ball and disappears. It will never exit the grid.



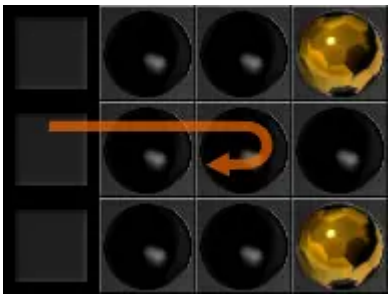
Rule 3: A ray cannot pass by a ball on its left or right. When a ray detects a ball that is one grid cell ahead and to the left or right, it will be deflected away from the ball at a right angle. Note that the deflection occurs *before* the ray comes up to the side of the ball. This distinction is what trips up most novice players.

Carefully observe the behavior in the diagram below; this is likely the most important concept that can be taken away from this guide.



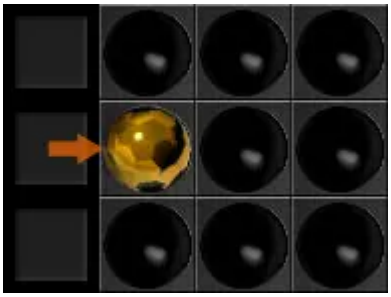
Rule 4: If a ray encounters two balls, one ahead and to the left, and one ahead and to the right, it does an *about face* and makes a 180° turn back the way it came.

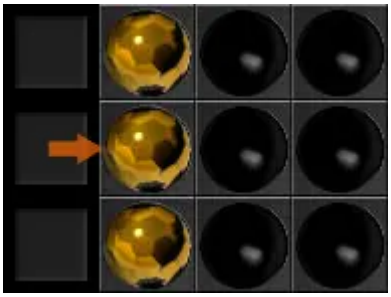
This behavior makes perfect sense, and is, in essence, a special case of Rule 3. The ball on the left turns the ray to the right, in which case the second ball, which is now to the turned ray's left, causes another turn to the right. This results in a complete 180° turn.



Rule 5: If a ball at the edge of the grid is to the ray's right or left, it prevents the ray from even entering the grid, and the ray appears to exit the grid at the entry port, resulting in a reflection.

Note that if there is a ball at the edge of the grid directly ahead of the ray, Rule 2 takes precedence regardless of the absence or presence of balls to the left or right, and the result is a hit. Both situations diagrammed below result in a hit.





Rule 2 always supersedes Rules 3 and 4, so if a ball is directly ahead of a ray, the result is always a hit.

Placing Guesses

Based upon the results obtained from shooting the rays, the player makes guesses as to the location of the balls.

Guesses are recorded by clicking on the cells within the grid. Clicking on an empty cell places a guess in that cell, indicated with a gold ball. Clicking on an occupied cell removes that guess.

Guesses can be freely placed and removed as many times as the player would like without penalty. There is no down-side to making, removing, or moving guesses around. In fact, it's an important part of visualizing the final solution.

Only as many guesses as there are balls hidden within the grid can be placed. So in a 4-ball game, up to 4 guesses can be placed. To move a guess, an existing guess must be removed before clicking on an empty cell to place a new guess.

Once the full number of guesses have been placed (3, 4 or 5 depending upon how many balls are selected at game start), these guesses can be submitted by clicking the **Submit Guesses** button.

This concludes the game and the results and final score are displayed.

Scoring

The final score is the summation of the amount of information needed to solve the puzzle, along with the accuracy of the guesses.

Lower scores are obviously better, and the numeric score value is computed as follows:

- One point for every **hit**.
- One point for every **reflection**.
- Two points for every **detour**.
- Five penalty points for every ball that is **guessed incorrectly**.

When the game is complete, the final score and how it was obtained is displayed in the control panel area, and the grid is updated to show the correct and incorrect guesses, as well as the actual locations of the balls.

Other Information

A new game can be started at any time by clicking the *Restart* button. If the current game is incomplete, the player will be prompted to confirm abandoning the running game.

Games can be bookmarked in the browser at any time, and returned to at a later date to resume playing the game.

Back New Game

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